



Impact of a Structured Transition Education Program on Transition Related Knowledge Among Adolescents with Congenital Heart Disease: A Single-Center Quality Improvement Study

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Abstract

As survival among individuals with congenital heart disease (CHD) improves, effective transition from pediatric to adult care has become increasingly important. We evaluated the impact of a structured transition education program on transition-related knowledge among adolescents with CHD. This single-center quality improvement study enrolled adolescents aged 12–18 years with CHD between February 2025 and April 2026. A baseline survey was administered to cardiology providers at our institution to characterize their current practices. Adolescent participants completed a 22-item Transition-related knowledge assessment survey adapted from the Self-Management and Transition to Adulthood with Treatment [Rx] Questionnaire (STARx) and TRxANSITION tools before and after a structured physician-led educational intervention. Education focused on five domains: CHD knowledge, medication management, healthcare navigation, reproductive health, and psychosocial misconceptions. Participants also received a personalized cardiac passport. Changes in correct responses were analyzed using McNemar's test. Fifty-three of 80 eligible adolescents participated (51% male; median age 15 years). Significant improvements in correct responses were observed in 19 of the 22 survey items, with absolute gains ranging from 21 to 76% points (all $p < 0.05$). The largest improvements occurred in knowledge of cardiac diagnosis and surgical history, medication management, health insurance navigation, and reproductive health. Baseline provider surveys revealed frequent counseling on disease-specific topics but limited discussion of reproductive health, insurance planning, and transition-related issues. A structured transition education intervention was associated with substantial improvements in disease knowledge, self-management skills, healthcare navigation, and reproductive health awareness among adolescents with CHD. These findings support integration of standardized transition education into routine pediatric cardiology practice.

Keywords Transition of care · Congenital heart disease · Transition education · Quality improvement

Abbreviations

CHD Congenital heart disease
ACHD Adult congenital heart disease
STARx Self-management and transition to adulthood

with Rx (Treatment) questionnaire
SQUIRE Standards for quality improvement reporting excellence

Introduction

Congenital heart disease (CHD) affects approximately 1% of live births, and more than 85% of affected children now survive into adulthood [1]. In 2010, an estimated 2.4 million individuals in the United States were living with CHD [2]. Adolescents and emerging adults of transition age are estimated to comprise 15% to 20% of the overall population with congenital heart disease [2]. Health care

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transition is the process of gaining independence with managing one's own medical care and is distinct from transfer of care which refers to the actual move from a pediatric to an adult provider [3]. The 2025 ACC/AHA/HRS/ISACHD/SCAI Guideline recommends that all patients with CHD should receive structured, patient-centered transition education that is age-, sex-, gender-, developmentally appropriate to improve knowledge and that CHD programs establish formal transfer-of-care policies for effective handoffs of patients from a pediatric to Adult congenital heart disease (ACHD) cardiologist [4].

Inadequate transition preparation has been associated with lapses in care, poorer disease knowledge, reduced self-management, increased emergency healthcare utilization, and delayed establishment of adult congenital heart disease care [5]. A meta-analysis of 17 studies encompassing 6,847 patients estimated a pooled discontinuity rate of 26.1% (95% CI 19.2–34.6%), with individual study estimates ranging from 8% to 76% depending on population and follow-up duration [6]. Patients receiving care in specialized ACHD programs have lower mortality than those without such access [7]. However, fewer than 30% of adults with complex CHD are enrolled with an ACHD provider [1]. Consequently, improving transition readiness has become a major quality priority in pediatric cardiology. Several randomized control studies support structured transition programs which consistently demonstrated improvement in CHD knowledge, health related quality of life, patient empowerment and reduce lapses in obtaining adult CHD care [3, 8, 9].

Despite the guideline recommendations, a minority of pediatric cardiology programs provide structured transition transfer to an ACHD provider with only 41% of 96 European CHD centers offering a structured transition program [10]. A multicenter US study of 401 adolescents and young adults with CHD across four tertiary centers found average knowledge deficits of 19% for medical topics, 19% for lifestyle topics, and 53% for pregnancy and family planning, confirming that recommended transition education is not available to most patients [11]. Although randomized trials have demonstrated the efficacy of structured transition programs, many interventions require dedicated personnel, multiple encounters, or substantial institutional resources. Data remain limited regarding the effectiveness of brief, clinic-based transition education interventions that can be integrated into routine pediatric cardiology visits. We implemented a structured transition education program for adolescents with CHD and evaluated its impact on transition-related knowledge using a survey adapted from validated transition assessment tools.

Materials and Methods

This is an ongoing single center quality improvement study conducted at Pediatric Cardiology clinic at University of Miami/Jackson Memorial hospital. The current report presents preliminary findings from participants enrolled between February 01, 2025, and April 30, 2026. This quality improvement initiative was reviewed and approved by the University of Miami Institutional Review Board and conducted using a prospective pre-post evaluation design. The study was reported in accordance with Standards for quality improvement reporting excellence (SQUIRE 2.0) guidelines. Participants 12–18 years of age with a diagnosis of CHD were included in this study. Participants were excluded if they had severe developmental delay impairing survey participation or a history of heart transplantation. Participants were recruited during their pediatric cardiology clinic visits. Informed consent was obtained from parents or the guardians, and assent was obtained from participants, as approved by the institutional review board.

Physicians and advanced practice providers who participated in patient care completed an initial survey regarding their transition of care practices with responses including (yes/no). A 22-item Transition-related knowledge assessment survey was developed to assess CHD knowledge across five domains: knowledge of CHD, medication knowledge, healthcare navigation skills, reproductive health knowledge, and psychosocial misconceptions. Items were derived from two validated transition instruments- the STARx questionnaire, which is an 18-item tool validated in adolescents with chronic conditions that identifies six factors accounting for approximately 65% of variance: medication management, provider communication, engagement during appointments, disease knowledge, adult health responsibilities, and resource utilization [12], and the TRxANSITION Scale, which is a 32-item tool that evaluates health care transition in 10 domains including knowledge about diagnosis, treatment, diet, reproductive health, school/work, insurance, ability to self-manage and identification of new health providers [13]. We retained the domain structure of these instruments - disease knowledge, medication management, provider communication and follow-up, insurance, reproductive health, and educational/vocational planning, while adapting item content to congenital heart disease. Modifications included new items on surgical history, endocarditis prophylaxis, and lifelong adult congenital cardiology follow-up; replacement of the nutrition domain with exercise and sports participation; and expanded reproductive health content covering pregnancy risk, medication teratogenicity, and contraception. Responses were simplified from graded scales to dichotomous self-report. The adapted survey was reviewed by members of the study team, including pediatric

cardiologists with expertise in transition of care, to establish face validity and ensure the relevance and clarity of survey items prior to implementation.

The transition education program consisted of 3 parts: (a) Transition-related knowledge assessment survey (Fig. 1) at the beginning of the clinic visit followed by (b) transition

education by the cardiologist and the principal investigator and finally (c) a repeat Transition-related knowledge assessment survey was done at the end of the clinic visit. During the clinic visit, prior to physician's clinical assessment, the participant completed the 22-item transition Transition-related knowledge assessment survey and another survey

Fig. 1 Transition-related knowledge assessment survey ^a

1. Do you know the name of your heart condition? Yes/no
2. Did you have a heart surgery? Yes/no
3. Do you know the name of your heart surgery? Yes/no
4. Do you know the names of your heart medications? Yes/no
5. Do you know what the medicines are for? Yes/no
6. Do you know what could happen if you stopped taking your medicines? Yes/no
7. Do you know how to refill your medication prescriptions? Yes/no
8. Do you know how to contact your doctor? Yes/no
9. Do you know how often you need to see a heart doctor? Yes/no
10. Do you know what symptoms or problems your doctor needs to know about? Yes/no
11. Do you know whether you will need to see a heart doctor for the rest of your life? Yes/no
12. Do you know whether you need antibiotics when you go to the dentist? Yes/no
13. Do you know what exercise or sports you can and cannot do? Yes/no
14. Do you know how to get health insurance when not covered by your parents' plan? Yes/no
15. Do you know whether your heart condition could affect your ability to get pregnant?
Yes/no
16. Do you know whether your heart condition could get worse if you got pregnant? Yes/no
17. Do you know whether any medicines could be harmful to an unborn baby? Yes/no
18. Do you know how someone with your heart condition can prevent unplanned pregnancy?
Yes/no
19. Because of my heart condition, I can't go to college. True/ False
20. Because of my heart condition, I can't get a job. True/ False
21. Because of my heart condition, I can't have a certain job. True/ False
22. Because of my heart condition, I can't have kids. True/ False

^aThe survey was administered before and after a structured, physician-led education session.

to assess their interest in various health-related topics using binary (yes/no) response options. The primary cardiologist and principal investigator taking into consideration the transition survey and participants interests conducted individualized transition-of-care education sessions with each participant. Transition education consisted of assessment of transition readiness, initiation of transition planning, education regarding cardiac diagnosis, surgical and procedural history, and medication management, counseling on infective endocarditis prophylaxis, symptom recognition, and exercise guidance, promotion of healthy dietary habits, reproductive, sexual health counseling, preparation for future health insurance, healthcare system navigation, assessment of needs related to peer support and mental health resources. The education session was further enhanced using a personalized heart passport (Fig. 2), with each session lasting approximately 30 min.

The transition-related knowledge assessment survey comprised 22 items across five domains. Items 1–18 used a yes/no response format, and items 19–22 used a true/false format. All survey items were scored dichotomously as correct or incorrect. The percentages of participants with correct responses before and after the educational intervention were compared for each survey item using McNemar's test for paired binary proportions, with 95% confidence intervals calculated using the Newcombe method. Items 4, 5, 6, 7 apply only to 32 participants who were on medications and questions 15–18 apply to 26 female participants. Participant educational interests are summarized descriptively and the proportion of participants expressing interest in each topic was reported as percentages. For subgroup analysis, participants were divided into younger group (12–14 years) and an older group (15–18 years). The percentage of items correctly answered across five transition-readiness domains were compared between two groups at baseline and post intervention using Mann-Whitney U test. Statistical significance was set at $p < 0.05$. Because analyses were exploratory and intended to identify educational domains most responsive to intervention, adjustment for multiple comparisons was not performed. The primary outcome was change in the proportion of correct responses for each survey item between pre- and post-intervention assessments. The secondary outcomes were domain-specific performance, participant educational interests, provider practices.

Results

Participant Characteristics

Between February 2025 and April 2026, 80 adolescents with congenital heart disease (CHD) met eligibility criteria,

of whom 53 (66.3%) completed both the pre- and post-education surveys and were included in the analysis (Fig. 3). Twenty-seven eligible patients (33.8%) did not participate. Documented reasons for non-participation included insufficient clinic time to complete the intervention ($n=2$), declined to participate ($n=2$), and clinic visit focused on counseling for an upcoming procedure ($n=4$); no reason was recorded for the remaining participants. The median age was 15 years (interquartile range [IQR] 13–16 years), with 22 participants (41.5%) aged 12–14 years and 31 (58.5%) aged 15–18 years. Twenty-seven participants (50.9%) were male. The cohort was racially diverse, with 60.8% identifying as White and 39.2% as Black; 54.9% identified as Hispanic. CHD complexity was categorized according to the ACC/AHA anatomic classification. Complex CHD was present in 36 participants (67.9%) including single-ventricle physiology, transposition of the great arteries, tetralogy of Fallot, and coarctation of the aorta; while 16 (30.2%) had simple CHD diagnoses such as atrial septal defect, ventricular septal defect, pulmonary valve stenosis, bicuspid aortic valve, coronary cameral fistula (Table 1).

Provider Transition Practices

Fourteen pediatric cardiology providers, including physicians and advanced practice providers, completed a baseline survey regarding transition-related counseling practices. Disease-specific education and medication adherence were addressed by all providers (14/14), while symptom recognition (13/14), physical activity recommendations (13/14), infective endocarditis prophylaxis (12/14), and sports participation guidance (12/14) were also commonly discussed. In contrast, transition-specific topics were addressed less consistently. Only 5 of 14 providers reported routinely discussing transition planning and dietary guidance, while reproductive health counseling and health insurance planning were each reported by only 2 of 14 providers.

Transition Related Knowledge Outcomes

Significant improvements in transition related knowledge domains were observed following the educational intervention (Table 2). Nineteen of 22 survey items demonstrated statistically significant improvement. The three items that did not achieve statistical significance had very high baseline correct-response rates, including knowledge of whether a prior cardiac surgery had been performed (95.9%) and rejection of the psychosocial misconceptions that CHD precludes attending college or obtaining employment (both 98.1%). In the CHD knowledge domain (items 1–3), the median percent correct rose from 39.6% at baseline to 95.9% following the intervention, an increase of 56.3%

Pediatric Cardiology – Health Passport:		Passport created date:								
Name of patient:	Date of birth:									
Primary Cardiologist:	Office phone number:									
Cardiology office address:										
Emergency contact:										
Cardiac diagnosis:	Heart diagram									
1.										
2.										
3.										
Surgical/ Cardiac history:										
1.										
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3.										
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Allergies:										
Medications:										
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Blood thinner:	INR goal:									
Contraception:										
Baseline oxygen saturation:	BP:	HR:								
Endocarditis prophylaxis before dental procedure: ____ Yes ____ No if yes: medication to be taken: ____										
History of arrhythmia: ____ AICD ____ Pacemaker										
Pacemaker dependent:										
Activity restriction ____ None ____ Avoid contact sports ____ Avoid weight lifting										
Activity recommended ____										
Any other recommendations not mentioned above:										
1.										
2.										

Fig. 2 Cardiac passport

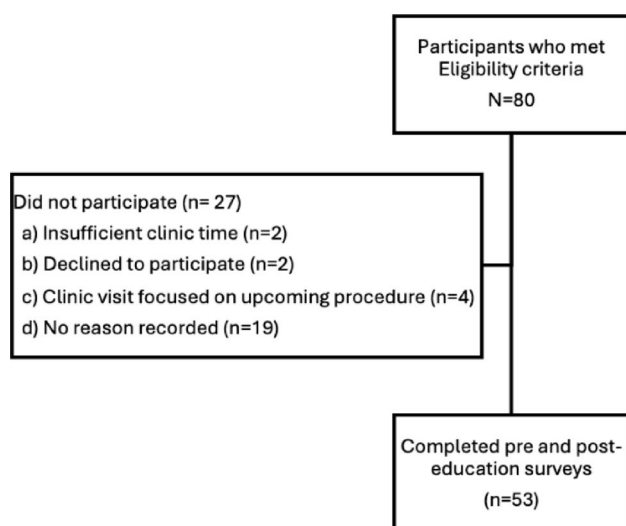


Fig. 3 Study participant selection

Table 1 Participant demographic and clinical characteristics ($N=53$)

Characteristics	$N=53$
Age	
Median age, years (IQR)	15 (13–16)
12–14 years	22 (41.5%)
15–18 years	31 (58.5%)
Sex	
Male	27 (50.9%)
Female	26 (49.1%)
Race	
White	32 (60.8%)
Black	21 (39.2%)
Ethnicity	
Hispanic	29 (54.9%)
CHD complexity	
Simple heart defects	16 (30.2%)
Complex heart defects	36 (67.9%)

points. The median percent correct in medication knowledge domain (items 4–7, $n=32$) improved from 46.9% to 81.3% (+34.4% points); in healthcare navigation skills (items 8–14), it increased from 55.8% to 90.6% (+34.8% points) and in Reproductive health knowledge domain (items 15–18, $n=26$), it improved from 18.4% to 91.9%, the largest gain observed in any domain (+73.5% points).

Age-Based Subgroup Analysis

At baseline the two age groups had no statistically significant difference across the domains. After the transition education intervention, performance remained statistically comparable between groups in four of the five domains including knowledge of CHD, medication knowledge, healthcare navigation, and reproductive health (all $p>0.28$). The single exception was the psychosocial-misconceptions domain, in

which the 15–18 group outscored the 12–14 group at post-test (99.2% vs. 92.0%; $p=0.012$). Overall, the intervention conferred comparable outcomes across the younger and older adolescent age groups (Table 3).

Adolescent Educational Interests

Participants identified several topics of interest for transition education (Table 4). The most frequently selected topic was understanding their cardiac diagnosis, surgeries, and procedures (88.7%), followed by future cardiology follow-up needs (77.4%), how to contact their cardiologist (75.5%), symptoms warranting medical attention (71.7%), and health insurance needs (71.7%). Interest was also common for exercise recommendations (62.3%), infective endocarditis prophylaxis (56.6%), and medication management (49.1%). Among female participants, 50.0% expressed interest in learning more about pregnancy and pregnancy prevention. Less frequently selected topics included vocational counseling (41.5%), stress management (30.8%), and community resources or counseling services (28.8%).

Discussion

In this single-center quality improvement initiative, implementation of a structured transition education program was associated with substantial improvements in transition-related knowledge among adolescents with congenital heart disease (CHD). Significant gains were observed across nearly all assessed domains, including disease-specific knowledge, medication management, healthcare navigation, reproductive health, and psychosocial understanding. These findings support the feasibility and effectiveness of incorporating standardized transition education into routine pediatric cardiology care.

Our findings are consistent with prior studies demonstrating that structured transition interventions improve disease transition related knowledge in adolescents with CHD. The CHAPTER 2 trial showed that a nurse-led educational intervention improved transition knowledge and reduced delays in establishing adult congenital heart disease (ACHD) care [7], while the TRANSITION-CHD [8] and STEPSTONES [9] randomized trials demonstrated improvements in disease knowledge, patient empowerment, and health-related quality of life. Collectively, these studies support the principle that transition readiness is a modifiable outcome that can be enhanced through targeted educational interventions. The present study extends this evidence by demonstrating that a practical clinic-based educational program delivered during routine outpatient visits can achieve meaningful

Table 2 Pre- and post-education correct response rates and McNemar test results ($N=53$)

Item	Survey question	Pre-survey (%) correct response	Post-survey (%) correct response	Difference in % correct responses (95% CI)	<i>p</i> -value
Knowledge of CHD diagnosis					
1	Knows name of heart condition	39.6	96.2	+56.6 (40.9 to 68.8)	<0.001
2	Knows whether they had a heart surgery and/ or procedure	95.9	95.9	0.0 (−7.7 to 7.7)	— ^a
3	Knows the name of heart surgery	33.3	89.6	+56.3 (39.5 to 68.4)	<0.001
Medication knowledge ($n=32$) ^b					
4	Knows names of cardiac medications	59.4	84.4	+25.0 (8.3 to 40.5)	0.005
5	Knows the purpose of the medication	46.9	84.4	+37.5 (16.1 to 54.6)	0.001
6	Knows consequences of stopping medications	46.9	71.9	+25.0 (6.1 to 41.2)	0.011
7	Knows how to refill medication prescriptions	40.6	78.1	+37.5 (16.1 to 54.2)	0.001
Healthcare navigation					
8	Knows how to contact the doctor	49.1	96.2	+47.2 (32.0 to 60.1)	<0.001
9	Knows follow-up frequency with cardiologist	75.5	96.2	+20.8 (9.2 to 33.4)	<0.001
10	Knows which symptoms to report	56.6	90.6	+34.0 (19.0 to 47.4)	<0.001
11	Knows lifelong cardiology care is needed	52.8	88.7	+35.8 (19.5 to 49.9)	<0.001
12	Knows about antibiotic prophylaxis before dental procedure	55.8	88.5	+32.7 (18.9 to 45.4)	<0.001
13	Knows exercise and sports restrictions	55.8	92.3	+36.5 (22.3 to 49.6)	<0.001
14	Knows how to obtain health insurance when not covered by parent's plan	5.8	61.5	+55.8 (39.9 to 67.9)	<0.001
Reproductive health (female participants $n=26$) ^c					
15	Knows CHD may affect ability to conceive	24.0	92.0	+68.0 (44.4 to 80.9)	<0.001
16	Knows pregnancy may worsen CHD	20.0	96.0	+76.0 (52.6 to 87.3)	<0.001
17	Knows the medications that may be harmful to fetus if they were to become pregnant	16.7	91.7	+75.0 (50.8 to 85.9)	<0.001
18	Knows contraceptive options	8.0	68.0	+60.0 (34.7 to 75.6)	<0.001
Psychosocial misconceptions (correct response = false)					
19	"I can't go to college because of my heart condition"	98.1	100	+1.9 (−5.0 to 9.9)	0.317
20	"I can't get a job because of my heart condition"	98.1	100	+1.9 (−5.0 to 9.9)	0.317
21	"I can't have a certain job because of my heart condition"	58.5	86.8	+28.3 (15.3 to 40.6)	<0.001
22	"I can't have kids because of my heart condition"	77.4	98.1	+20.8 (9.1 to 33.6)	<0.001

CI Confidence interval, CHD congenital heart disease

^a Not computable owing to insufficient discordant pairs (ceiling effect)

^b Items 4–7 administered only to participants prescribed with cardiac medications

^c Items 15–18 administered only to female participants

improvements in knowledge acquisition without requiring extensive additional resources.

The greatest improvements occurred in areas where baseline knowledge was poorest. Notably, fewer than half of participants could identify their cardiac diagnosis or surgical history before the intervention. This finding is concerning, as understanding one's underlying cardiac condition is a fundamental component of successful transition and self-management. Following education, most participants were able to identify their diagnosis and prior procedures, suggesting

that these knowledge deficits are readily addressable when information is presented in a structured and individualized manner. The use of a personalized cardiac passport likely contributed to these gains by providing participants with a portable and accessible summary of their medical history.

A particularly important finding was the improvement in healthcare navigation skills. Adolescents demonstrated limited baseline knowledge regarding how to contact their cardiology team, recognize symptoms requiring medical attention, and navigate future healthcare needs. Most

Table 3 Age-based subgroup analysis of transition related knowledge domains before and after the intervention

Domain	Assessment	12–14 years % correct responses (n)	15–18 years % correct responses (n)	p-value
Knowledge of CHD	Baseline	53.2 (22)	58.0 (31)	0.377
	Post-intervention	93.5 (22)	94.3 (31)	0.755
Medication knowledge	Baseline	46.2 (13)	50.0 (19)	0.681
	Post-intervention	76.9 (13)	81.2 (20)	0.355
Healthcare navigation skills	Baseline	52.6 (22)	48.8 (31)	0.500
	Post-intervention	87.4 (22)	88.0 (31)	0.497
Reproductive health knowledge	Baseline	20.8 (12)	18.2 (14)	0.977
	Post-intervention	79.5 (11)	92.7 (14)	0.286
Psychosocial misconceptions	Baseline	83.0 (22)	83.1 (31)	0.874
	Post-intervention	92.0 (22)	99.2 (31)	0.012

Table 4 Adolescent-identified educational interests (N=53)

Items	Topic	Interested, % (n/N)
1	Your heart condition, surgery and procedures	88.7 (47/53)
2	Future needs for follow-up visits	77.4 (41/53)
3	How to contact your heart doctor	75.5 (40/53)
4	Symptoms warranting a call to your doctor	71.7 (38/53)
5	Health insurance needs	71.7 (38/53)
6	Exercise, sports recommendations	62.3 (33/53)
7	Antibiotic prophylaxis for dental procedures	56.6 (30/53)
8	Pregnancy / pregnancy prevention ^a	50% (13/26)
9	Your medications	49.1 (26/53)
10	Job or vocational counseling	41.5 (22/53)
11	Stress management	30.8 (16/52)
12	Community resources, including counseling	28.8 (15/52)

Interest was assessed at baseline using a binary (yes/no) format. N=53 for all items except stress management and community resources (n=52; one missing response for each)

^a the pregnancy / pregnancy-prevention- item was most relevant to the 26 female participants

strikingly, knowledge regarding acquisition of health insurance after aging out of parental coverage was extremely poor at baseline yet improved substantially following education. Insurance instability and loss of follow-up remain important contributors to care discontinuity during the transition period. Given that approximately one-quarter of young adults with CHD experience interruptions in cardiac care, improving understanding of healthcare access and insurance planning may represent a critical component of successful transition programs.

The reproductive health findings warrant special attention. Female participants demonstrated profound baseline knowledge deficits regarding pregnancy-related cardiovascular risks, potential teratogenicity of cardiac medications, and contraceptive considerations. These observations mirror findings from multicenter studies showing that reproductive health represents one of the largest unmet educational needs among adolescents and young adults with CHD. Despite guideline recommendations emphasizing preconception

counseling and reproductive health education, these topics remain inconsistently addressed in routine pediatric cardiology practice. Encouragingly, reproductive health knowledge showed the largest gains following the intervention, indicating that these discussions are both feasible and effective when incorporated into structured transition programming. Nevertheless, post-intervention knowledge regarding contraception remained lower than for other reproductive health topics, suggesting that this area may require repeated counseling and collaboration with adolescent medicine and gynecology specialists.

An important strength of this study is the integration of provider perspectives and patient-reported educational interests. The provider survey revealed that clinicians consistently addressed diagnosis-specific topics, medication adherence, and symptom recognition but were less likely to discuss insurance planning, reproductive health, and other transition-specific concerns. Notably, these same domains corresponded to the largest patient knowledge deficits. Furthermore, adolescents expressed substantial interest in learning about health insurance, future healthcare needs, and management of their cardiac condition. The alignment between provider omissions, patient knowledge gaps, and patient-identified educational priorities highlights a clear opportunity for improvement. Structured transition curricula may help standardize delivery of essential content and reduce variability in counseling practices across providers.

The subgroup analysis demonstrated broadly similar patterns of improvement among younger and older adolescents, suggesting that transition education can be effectively introduced as early as 12 years of age. Current transition guidelines emphasize a longitudinal process that begins in early adolescence rather than a single intervention immediately before transfer to adult care. Our findings support this approach and suggest that foundational concepts can be successfully introduced well before the anticipated transfer to an Adult congenital heart disease specialist.

These findings have important implications for pediatric cardiology programs. Despite increasing recognition of

transition as a core component of CHD care, implementation of structured transition programs remains inconsistent across institutions. The present initiative demonstrates that a standardized educational intervention can be integrated into routine clinic workflows and can address critical knowledge gaps that may otherwise persist into adulthood. Incorporating tools such as personalized cardiac passports, structured educational checklists, and dedicated transition visits may facilitate broader implementation and improve consistency of care delivery.

Unlike many previously studied transition interventions that relied on dedicated nurse coordinators or multiple structured sessions, the present intervention was integrated into routine outpatient encounters and delivered by existing clinical personnel. This approach may enhance scalability and facilitate adoption in resource-limited pediatric cardiology programs.

Limitations

Several limitations warrant consideration. First, the single-group pre-post design precludes attribution of improvements solely to the intervention and introduces potential testing and Hawthorne effects. Second, knowledge was assessed immediately after education and therefore reflects short-term knowledge acquisition rather than durable learning. Third, the survey instrument was adapted from validated transition tools but has not undergone formal psychometric validation in adolescents with CHD. Fourth, the single-center design and modest sample size may limit generalizability. Fifth, participation was voluntary and one-third of eligible patients did not participate, introducing potential selection bias. Finally, improvements in knowledge may not translate into behavioral outcomes such as self-management, continuity of care, or successful transfer to ACHD services. Future work should evaluate long-term retention of transition knowledge and examine whether educational gains translate into improved self-management, reduced lapses in care, and successful establishment of ACHD follow-up.

Conclusion

A structured transition education program was associated with significant improvements in disease-specific knowledge, self-management skills, healthcare navigation, and reproductive health awareness among adolescents with CHD. The largest gains occurred in domains that were infrequently addressed during routine clinical encounters yet were highly valued by patients. These findings support the integration of standardized transition education into

pediatric cardiology practice and underscore the importance of addressing both medical and nonmedical aspects of transition preparation, including healthcare navigation, reproductive health, and insurance planning.

Author Contributions Dr. Keerthana Banala has conceptualized and designed the study, collected data, and drafted the initial manuscript- Leonardo Morales has assisted with designing the study and with data collection - Dr. Nelson Claure has carried out the statistical analyses and interpreted the data- Dr. Satinder Sandhu has conceptualized and designed the study, supervised the project and critically reviewed the manuscript.- All authors reviewed and revised the manuscript, approved the -final version as submitted, and agree to be accountable for all aspects of the work.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interests.

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